DESCRIPTIVE

ANALYTICS

LAY'S CHIPS

SALES • CUSTOMER PREFERENCES • PRODUCT PERFORMANCE

PROJECT REPORT

ACADEMIC SESSION

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Course	Bachelor of Computer Applications (BCA) — DS & AI
Subject	Descriptive Analytics / Data Visualization
Dataset	Synthetic LAY'S Chips Sales Dataset
Analysis Period	1 July 2026 – 30 July 2026

DATA-DRIVEN SNACK-FOOD INSIGHTS

Academic disclaimer: All transaction-level findings are generated from a synthetic dataset for educational purposes. They are not official Lay's / PepsiCo sales, customer, revenue, market-share, or product-performance statistics.

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Keywords

Keywords: LAY'S Chips, Descriptive Analytics, Data Visualization, Sales Analysis, Customer Preferences, Product Performance, Snack-Food Analytics, Synthetic Dataset, Python, Pandas, Jupyter Notebook, Matplotlib, Data Cleaning, Sales Trend, Customer Ratings, Flavour Categories, Pack Sizes, Flavour Preference, City-wise Sales, Online Sales, Offline Sales, Discounts, Return Behaviour, Key Performance Indicators (KPIs), Business Insights, Predictive Analytics, Retail Analytics, Inventory Planning, Customer Segmentation.

Key Terms Used in the Report

• **Descriptive Analytics:** The process of organizing, summarizing and interpreting historical or observed data to understand what happened.

• **Data Visualization:** Representing numerical information through charts and graphs to make trends, comparisons and patterns easier to identify.

• **Product Performance:** A measure of how individual snack products perform using indicators such as units sold, revenue and customer ratings.

• **Customer Preference:** The observed tendency of customers toward particular products, pack sizes, flavours, categories, channels or price levels.

• **Synthetic Dataset:** Artificially generated transaction data designed to resemble realistic records for academic analysis.

• **Business Insight:** A practical interpretation of an analytical result that can support inventory, marketing, merchandising or customer-experience decisions.

• **KPI:** A key performance indicator that summarizes an important business measure.

• **Predictive Analytics:** A future analytical stage in which historical data are used to estimate future demand or customer behaviour.

Analytical Scope

The report focuses on packaged-snack sales and customer behaviour. The analytical approach follows the sample report by presenting the figure context, observation, business insight and measurable KPIs for each major topic, while replacing the original subject completely with LAY'S chips.

Note on Data

This report uses a synthetic LAY'S chips sales dataset created specifically for academic descriptive analytics. It is not a company-owned or officially published Lay's / PepsiCo transaction database.

The dataset contains 150 transaction records representing a simulated 30-day period from 1 July 2026 to 30 July 2026. It includes product information, customer attributes, pricing, discounts, sales quantities, revenue, location, channel, payment method and return status.

The dataset contains multiple simulated flavour products and Indian locations so that the project can demonstrate product-level, customer-level, regional and channel-level analysis. Because the records are synthetic, the resulting values should be used to demonstrate the analytical process rather than to make claims about actual company performance.

Dataset Variables

Product_ID — Unique product identifier
Product_Name — Snack flavour/product name used for product ranking
Category — Flavour category such as Classic, Spicy, Tangy, Smoky, Cheesy and Baked
Pack_Size — Product pack size used for segmentation
Price — Listed product price
Discount — Discount percentage applied to the listed price
Final_Price — Price after discount
Units_Sold — Number of units represented by a transaction
Revenue — Final_Price multiplied by Units_Sold
Customer_Age — Simulated customer age
Customer_Rating — Simulated rating from 1 to 5
Purchase_Date — Date of simulated purchase
City — Simulated market location
Sales_Channel — Online or Offline/Store
Payment_Method — Payment mode
Return_Status — Returned or not returned

Objectives

The main purpose of this descriptive analytics project is to understand the simulated sales performance and customer preferences associated with LAY'S chips. The project converts raw transaction records into understandable charts, summary measures and business insights.

- Identify the most purchased LAY'S products and compare their sales performance.
- Analyse daily sales activity across the 30-day simulated period and identify high- and low-sales dates.
- Identify peak purchase hours and understand when customers are most active.
- Compare the popularity of Classic, Spicy, Tangy, Smoky, Cheesy and Baked flavour categories.
- Measure average customer ratings and identify stronger and weaker-rated categories.
- Compare simulated chips sales across major Indian cities and markets.
- Compare Online and Offline/Store sales to understand channel contribution.
- Analyse customer age groups and identify the segment contributing the greatest number of units.
- Identify the most demanded pack sizes and flavours for inventory planning.
- Compare sales across discount ranges without incorrectly claiming causation.
- Analyse return behaviour and identify categories requiring additional investigation.
- Calculate KPIs that summarize product, customer, regional and channel performance.
- Explain how the same descriptive framework could support inventory, marketing and merchandising at real scale.
- Establish a foundation for future predictive analytics such as demand forecasting and return prediction.

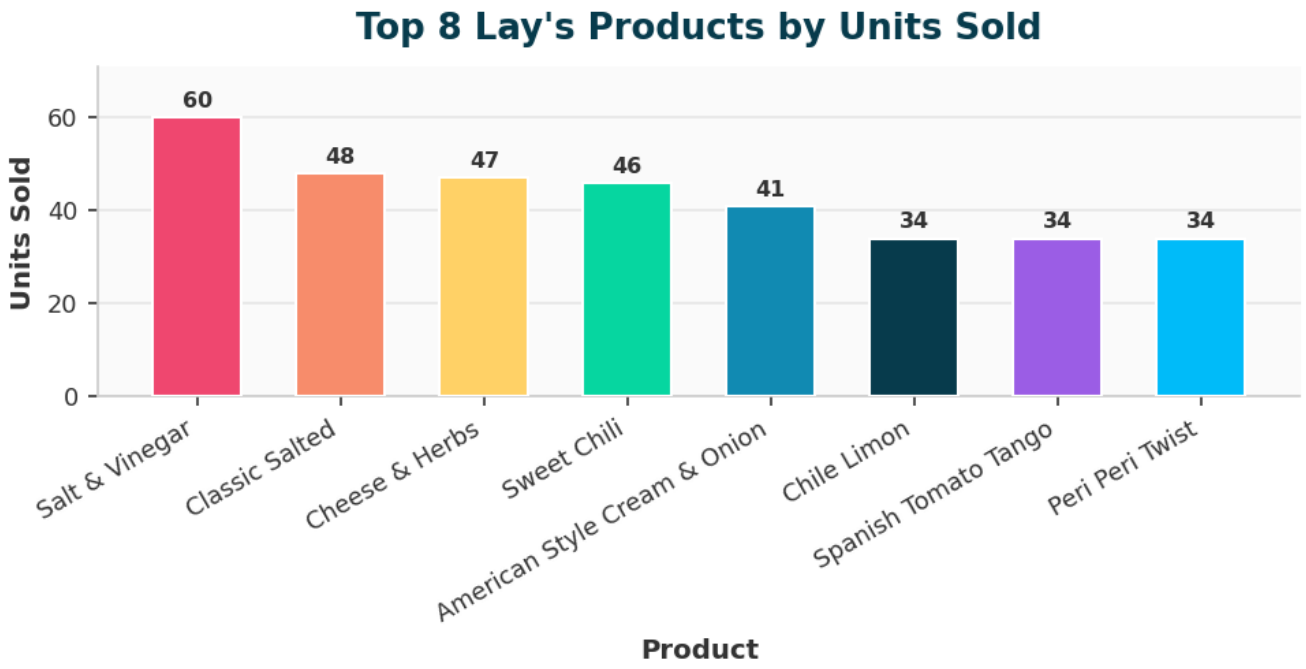
Expected Outcome

The expected outcome is a structured analytical report in which every major visualization is supported by a clear description, observation and business insight. This makes the report suitable for a BCA/Data Visualization academic submission.

1. Most Purchased LAY'S Products

Figure Description

The figure presents the ranking of LAY'S products according to total simulated Units_Sold during the 30-day analysis period. The X-axis represents the individual products, while the Y-axis represents the total number of units sold. The bars allow direct comparison between the highest-demand products and products with comparatively lower sales. The visualization is useful for identifying products that may require greater inventory attention.



KPI

- Best-selling product: Salt & Vinegar — 60 units
- Second-highest product: Classic Salted — 48 units
- Third-highest product: Cheese & Herbs — 47 units
- Total simulated units sold: 440

Observation

The graph indicates that Salt & Vinegar recorded the highest number of simulated units sold. Classic Salted and Cheese & Herbs also showed strong demand, while the lower-ranked products generated comparatively fewer units. The difference between the leading products and the remaining products suggests that customer demand is not evenly distributed across the complete product range. This creates a clear product-performance hierarchy that can be used for further analysis.

Business Insight

High-performing products should receive priority in inventory replenishment, website visibility and store merchandising. Featured products can be supported with promotional campaigns when appropriate, while lower-performing products should be reviewed for price, flavour positioning, pack availability, customer ratings and shelf placement. In a real retail environment, product-level demand would also be combined with profit margin and stock availability before making final decisions.

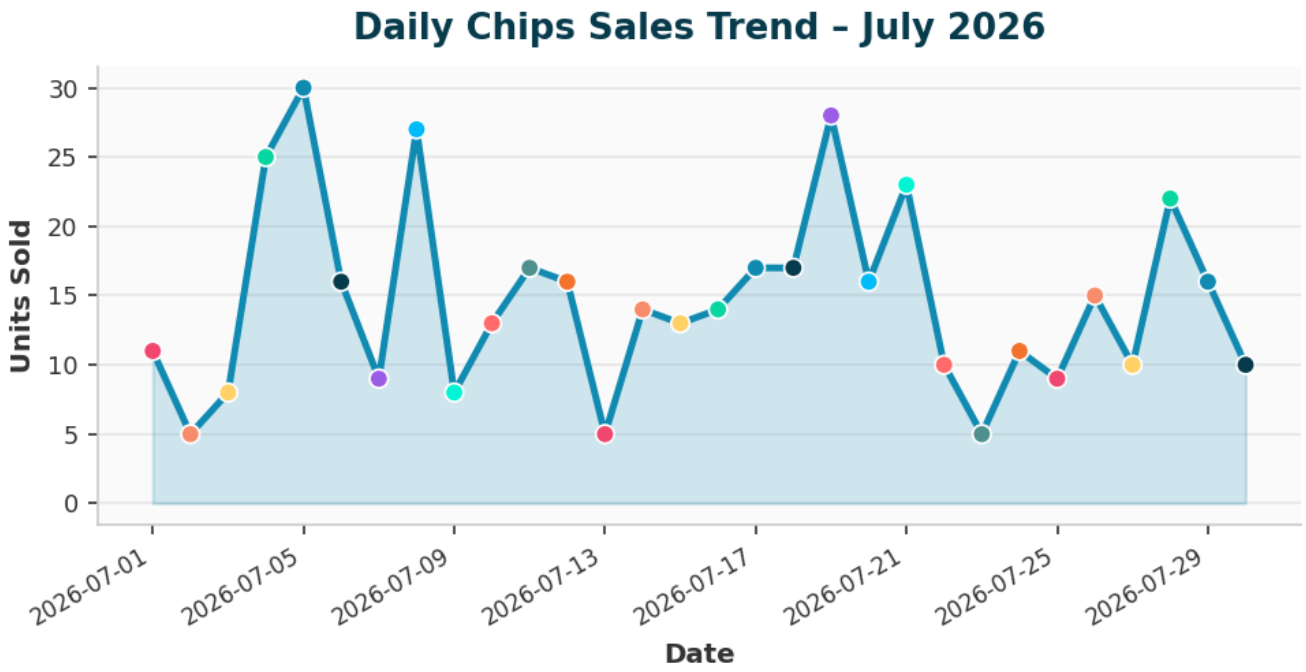
Additional Interpretation

Product rankings alone do not explain why a product performs well. A stronger analysis would combine product demand with price, discount, rating, return rate, pack availability and category.

2. Daily Sales Trend

Figure Description

The figure shows the Daily Sales Trend of simulated chips units sold over the 30-day analysis period from 1 July 2026 to 30 July 2026. The X-axis represents the Purchase_Date and the Y-axis represents total Units_Sold for each day. Each point represents the aggregated units sold on that date, and the connected line shows how sales activity changed across the month. The chart helps identify sales peaks, lower-volume days and general fluctuations in daily demand.



KPI

- Average daily units sold: 15
- Highest-sales date: 2026-07-05 — 30 units
- Lowest-sales date: 2026-07-02 — 5 units

Observation

The graph shows noticeable fluctuations in daily sales rather than a perfectly constant pattern. The highest simulated activity occurred on 2026-07-05, while 2026-07-02 recorded the lowest value. Other peaks and dips appear across the month, indicating that daily purchase activity changes over time. In real retail data, such changes could be associated with weekends, promotions, product availability, salary cycles, holidays or other external factors.

Business Insight

Identifying high-sales days can help businesses prepare inventory and schedule marketing campaigns more effectively. Additional stock can be positioned before expected high-demand periods, while promotional activity can be timed to increase traffic during strategically important dates. Daily monitoring can also help identify unusual drops that may indicate stock-outs, technical problems or weaker customer demand.

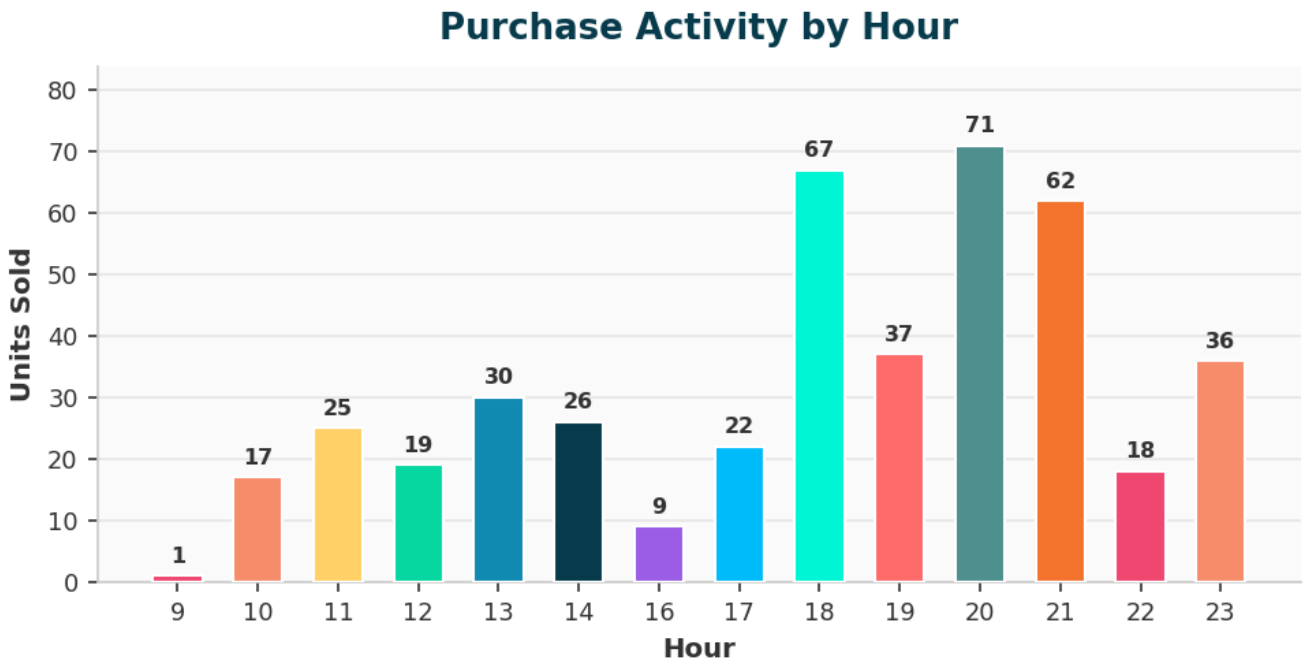
Additional Interpretation

The 30-day synthetic period demonstrates the analytical method but should not be interpreted as evidence of actual seasonal behaviour.

3. Peak Purchase Hours

Figure Description

The figure presents the distribution of simulated Units_Sold by Purchase_Hour. The X-axis represents the hour of purchase and the Y-axis represents total units sold during that hour. Comparing hourly activity makes it possible to identify the periods when customers are most active. This type of analysis is useful for online merchandising, promotional scheduling, customer support and operational planning.



KPI

- Peak purchase hour: 20:00
- Units sold at peak hour: 71
- Lowest purchase hour: 9:00 — 1 units

Observation

The graph indicates that purchase activity varies by time of day. The highest simulated activity is concentrated around 20:00, while 9:00 records the lowest activity. The pattern suggests that customer purchasing behaviour is stronger during selected hours rather than being evenly distributed across the day.

Business Insight

Validated peak-hour information can help retailers schedule digital promotions, notifications and customer-support resources when customers are most active. Online platforms can highlight selected flavours before peak periods, while order-processing and service teams can prepare for increased activity. Real store footfall data could similarly support staff scheduling.

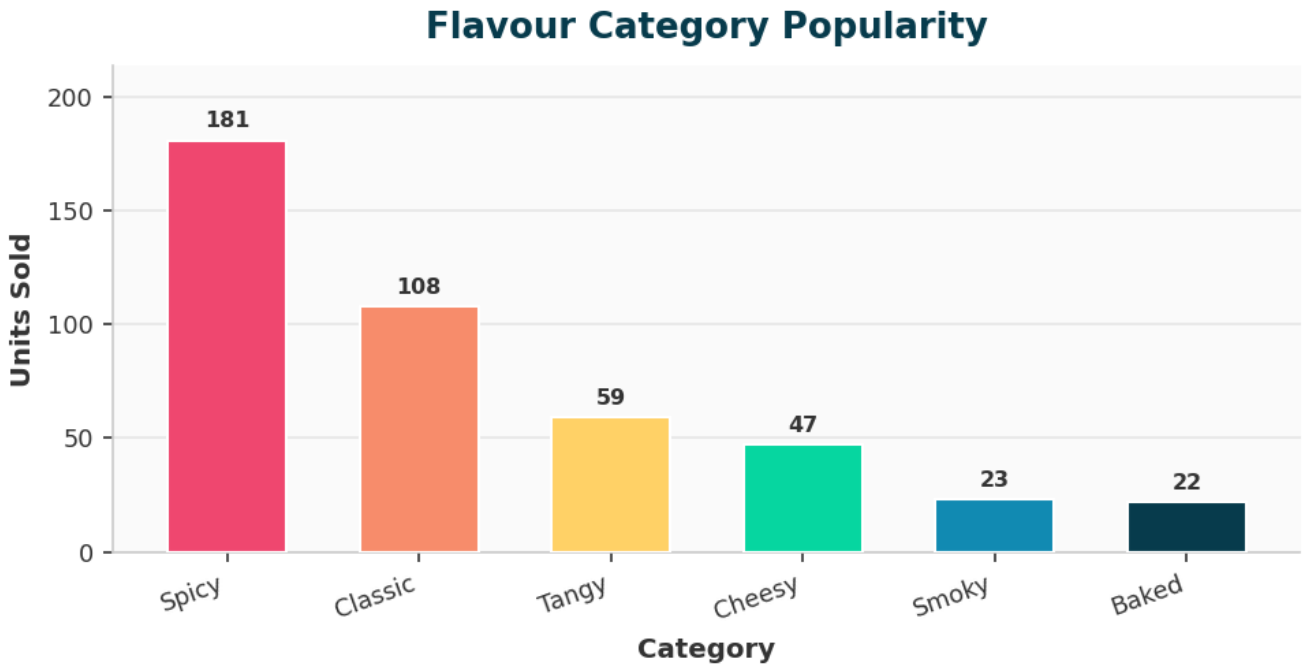
Additional Interpretation

Peak-hour analysis is descriptive; it identifies when activity occurs but does not by itself explain the reason for that activity.

4. Flavour Category Popularity

Figure Description

The figure compares simulated units sold across major flavour categories including Classic, Spicy, Tangy, Smoky, Cheesy and Baked. The X-axis represents categories and the Y-axis represents total Units_Sold. The chart shows which categories contributed larger or smaller portions of total chips demand during the simulated period.



KPI

- Best-selling category: Spicy — 181 units
- Second-highest category: Classic — 108 units
- Total categories analysed: 6

Observation

Spicy is the leading simulated flavour category, followed by Classic and other categories. The variation between categories indicates that customer demand differs according to flavour type. Categories with stronger sales contribute a larger share of total units and may represent higher customer interest within this synthetic dataset.

Business Insight

Popular categories should receive greater inventory attention and merchandising visibility. Lower-performing categories can be examined for pricing, flavour intensity, availability and promotional effectiveness. Category-level decisions should also consider revenue and margin because a category with fewer units may still be commercially valuable if its average selling price or margin is higher.

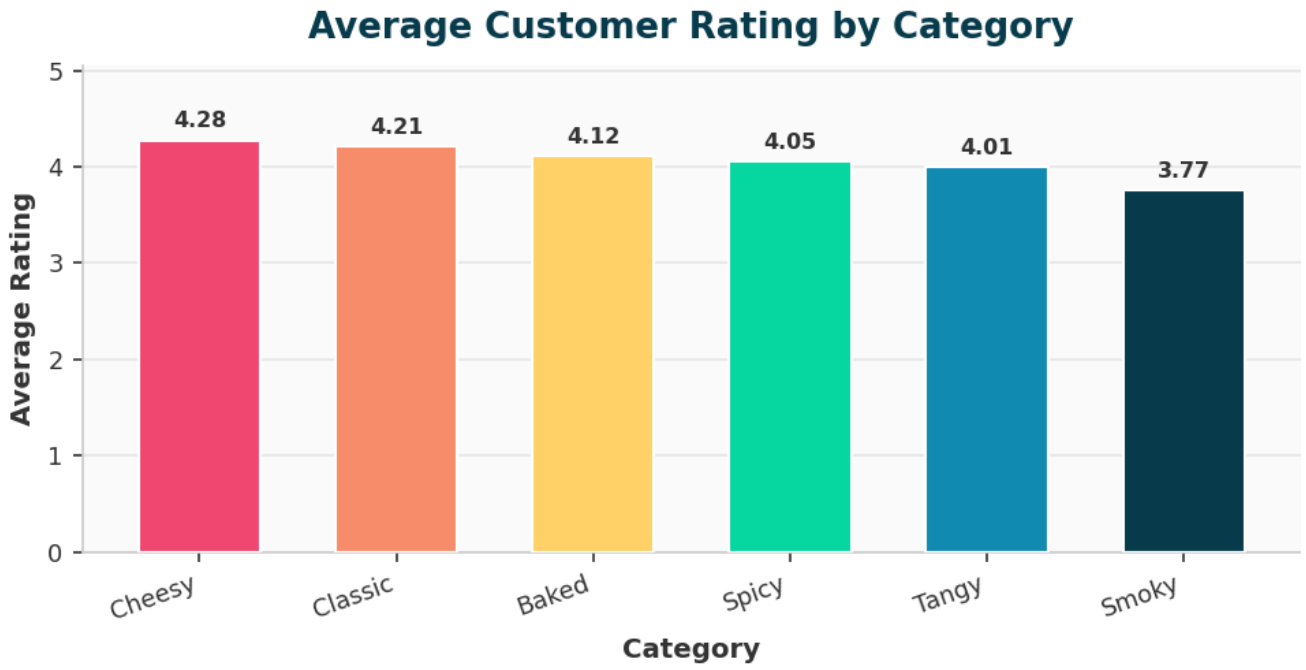
Additional Interpretation

Category popularity is a descriptive measure and should be combined with profitability, return rates and stock levels for real business decisions.

5. Customer Ratings

Figure Description

The figure presents the Average Customer Rating by flavour category. The X-axis represents the categories and the Y-axis represents average rating on a 1–5 scale. The purpose is to compare customer satisfaction across product groups and identify categories with relatively stronger or weaker ratings.



KPI

- Highest-rated category: Cheesy — 4.28/5
- Lowest-rated category: Smoky — 3.77/5
- Overall average rating: 4.09/5

Observation

The ratings show differences between flavour categories. Cheesy has the highest simulated average rating of 4.28, while Smoky has the lowest average at 3.77. The differences can help identify categories that appear to provide stronger customer satisfaction in the academic dataset.

Business Insight

Highly rated categories can be highlighted in product recommendations and merchandising. Categories with lower ratings should be investigated through review comments and return reasons to identify possible concerns related to taste, freshness, packaging or spice-level expectations. Improving product information may also help set customer expectations.

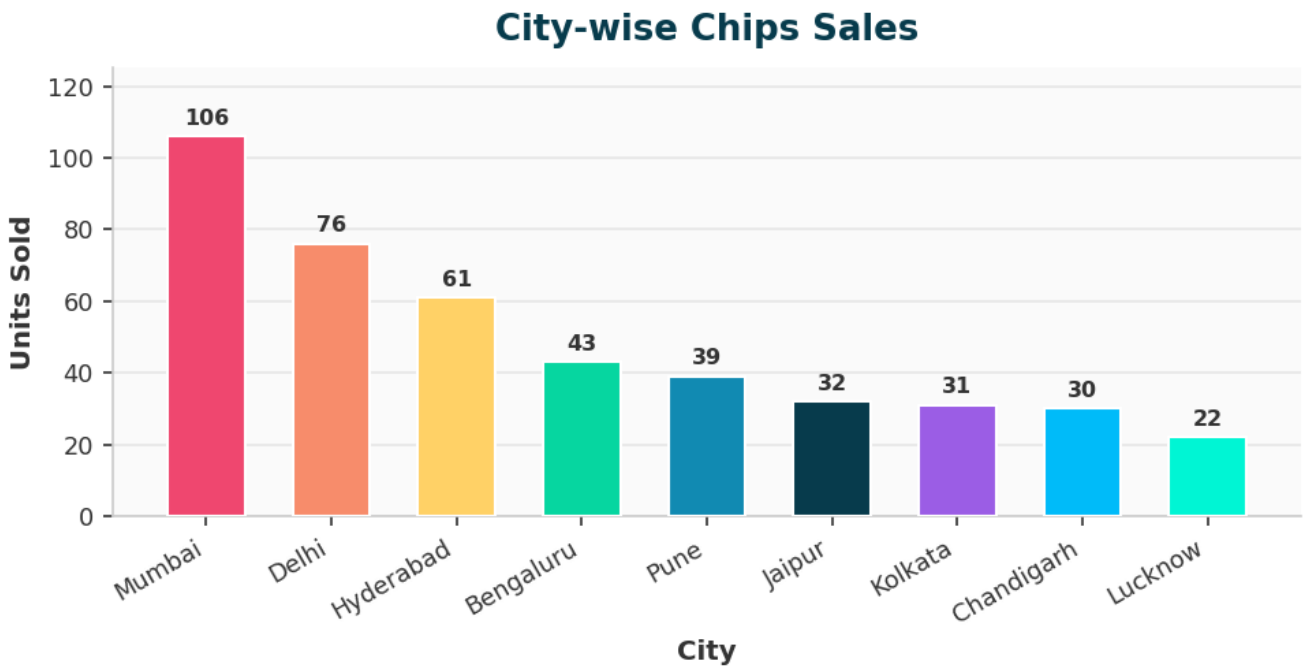
Additional Interpretation

Ratings in this project are synthetic and should not be presented as official Lay's / PepsiCo customer-review statistics.

6. City-wise Sales

Figure Description

The figure compares simulated Units_Sold across the Indian cities represented in the dataset. The X-axis represents cities and the Y-axis represents total units sold. The chart is intended to show regional concentration of demand and identify markets that contribute comparatively larger or smaller sales volumes.



KPI

- Highest-sales city: Mumbai — 106 units
- Second-highest city: Delhi — 76 units
- Cities represented: 9

Observation

The graph indicates that Mumbai contributes the highest simulated sales volume, followed by Delhi and other cities. Other cities show comparatively smaller unit volumes. The variation suggests that customer demand is geographically uneven within the synthetic records.

Business Insight

High-performing cities could receive greater inventory depth and targeted promotional campaigns when supported by real sales evidence. Flavour and pack-size preferences can be analysed within each city to improve local assortment. Regional distribution planning can also reduce the risk of stock-outs in stronger markets.

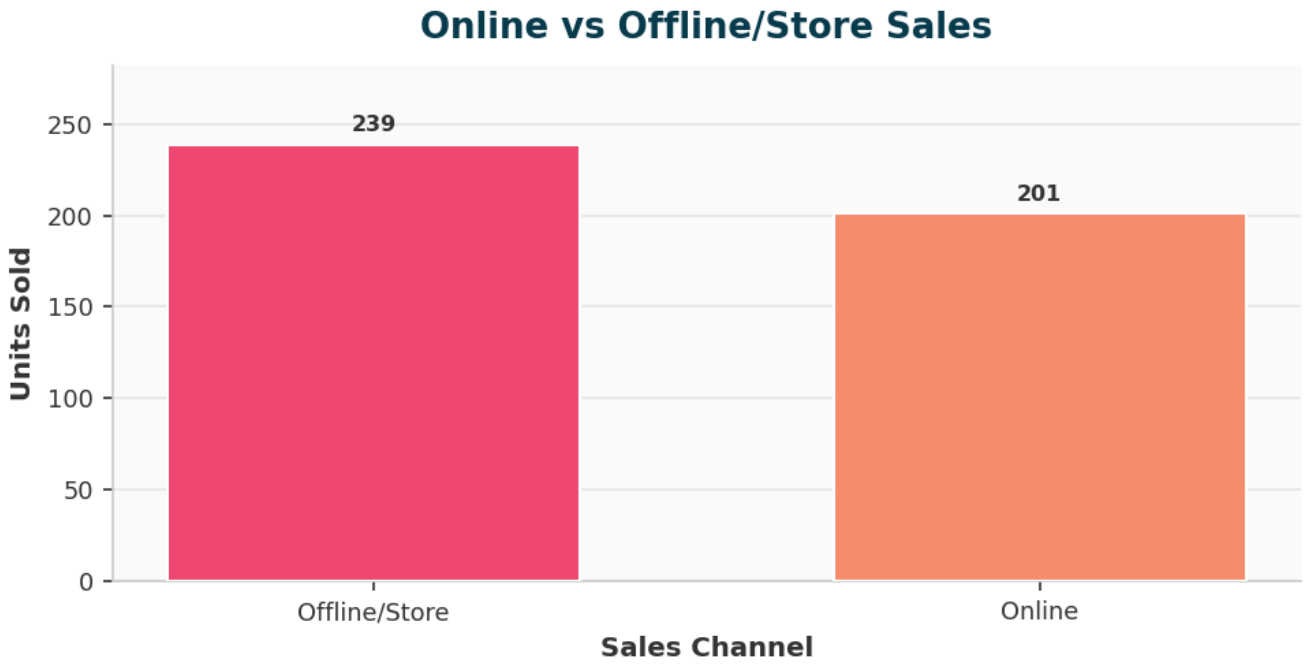
Additional Interpretation

These results are synthetic and must not be interpreted as official Lay's city-wise or state-wise sales.

7. Online vs Offline Sales

Figure Description

The figure compares sales performance across the simulated Online and Offline/Store channels. The analysis considers both Units_Sold and Revenue so that channel contribution is not judged only by transaction volume. Comparing channels helps identify where customers are purchasing and where additional merchandising or operational attention may be useful.



KPI

- Leading channel by units: Offline/Store — 239 units
- Online revenue: Rs. 11,078.0
- Total channel revenue: Rs. 24,729.5

Observation

The channel comparison shows differences between Online and Offline/Store activity. Offline/Store contributes the larger number of simulated units sold. The revenue comparison provides an additional view because a channel may sell fewer units but generate substantial revenue depending on pack sizes and prices.

Business Insight

Channel performance can guide marketing allocation, inventory placement and customer-experience improvements. Strong online demand can justify stronger digital merchandising and availability, while strong store demand can support local inventory depth and in-store promotions. A unified inventory view can help avoid selling popular flavours online while stores face shortages.

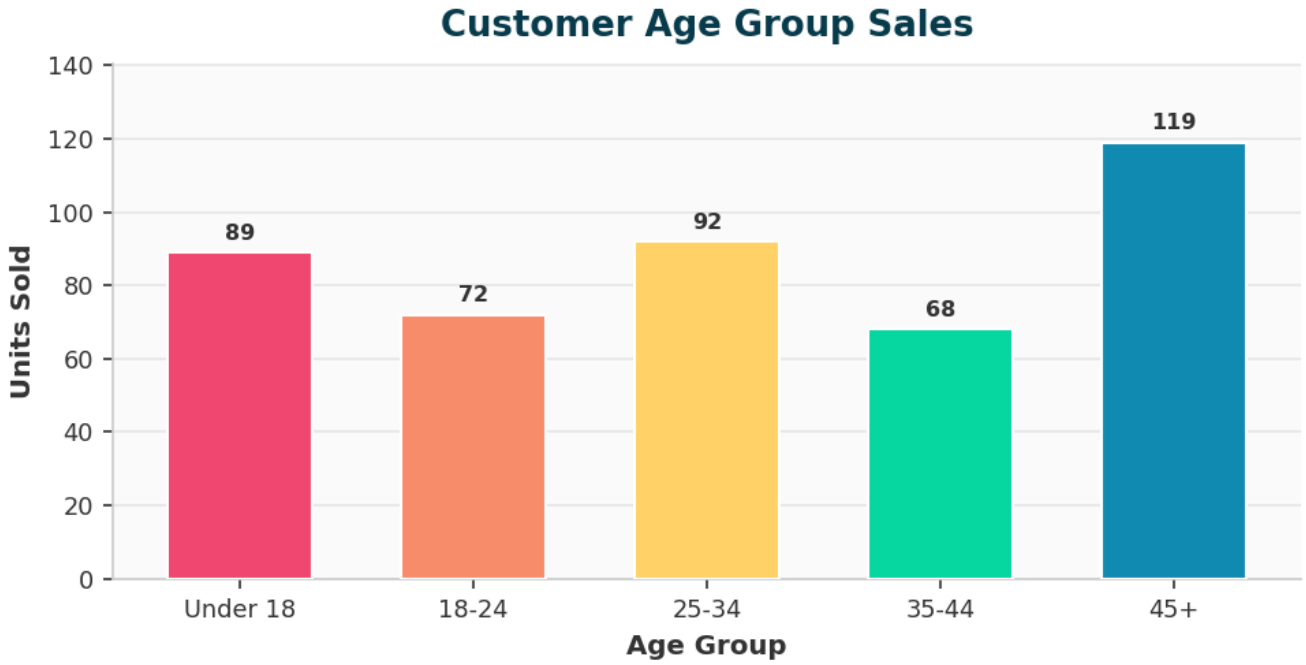
Additional Interpretation

Channel results may be affected by the synthetic sampling design and should be validated with actual sales and margin data.

8. Customer Age Preferences

Figure Description

The figure presents simulated chips purchases across five customer age groups: Under 18, 18–24, 25–34, 35–44 and 45+. The X-axis represents the age group and the Y-axis represents total Units_Sold. Age segmentation provides a simple view of which customer segments contribute more purchase volume.



KPI

- Highest-volume age group: 45+ — 119 units
- Lowest-volume age group: 35-44 — 68 units
- Age groups analysed: 5

Observation

The age-group comparison shows that 45+ contributes the highest simulated unit sales. Other groups contribute different levels of demand, indicating that purchase volume is not equally distributed across all age segments.

Business Insight

The strongest age segment can be used as a starting point for targeted campaigns, product recommendations and communication. However, businesses should avoid ignoring smaller segments; targeted offers and suitable pack-size assortments can help improve engagement across customer groups.

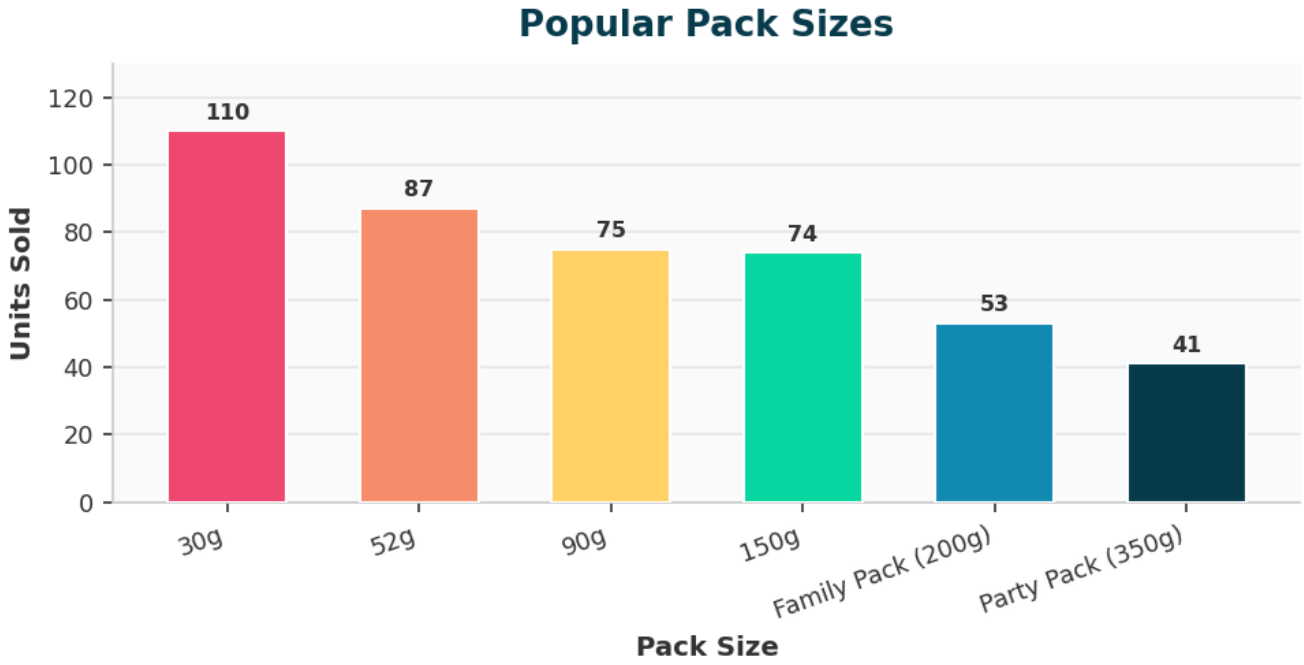
Additional Interpretation

Age-group results are simulated and do not represent official Lay's customer demographics.

9. Pack Size Preference

Figure Description

The figure presents total simulated Units_Sold by pack size. The X-axis represents available pack sizes and the Y-axis represents the number of units sold. Pack-size analysis is important for packaged snacks because assortment availability by size directly affects whether a customer can complete a purchase.



KPI

- Most demanded pack size: 30g — 110 units
- Second-most demanded pack size: 52g — 87 units
- Pack sizes analysed: 6

Observation

The 30g pack records the highest simulated demand, while other pack sizes contribute different levels of sales. The chart highlights the relative demand distribution across the available pack sizes and can help identify sizes that require greater attention.

Business Insight

Popular pack sizes should be maintained at sufficient stock levels to reduce stock-outs. Real retailers can combine pack-size demand with product, city and category to determine exactly where additional inventory is required. Impulse-purchase small packs and value-oriented large packs may need different placement strategies.

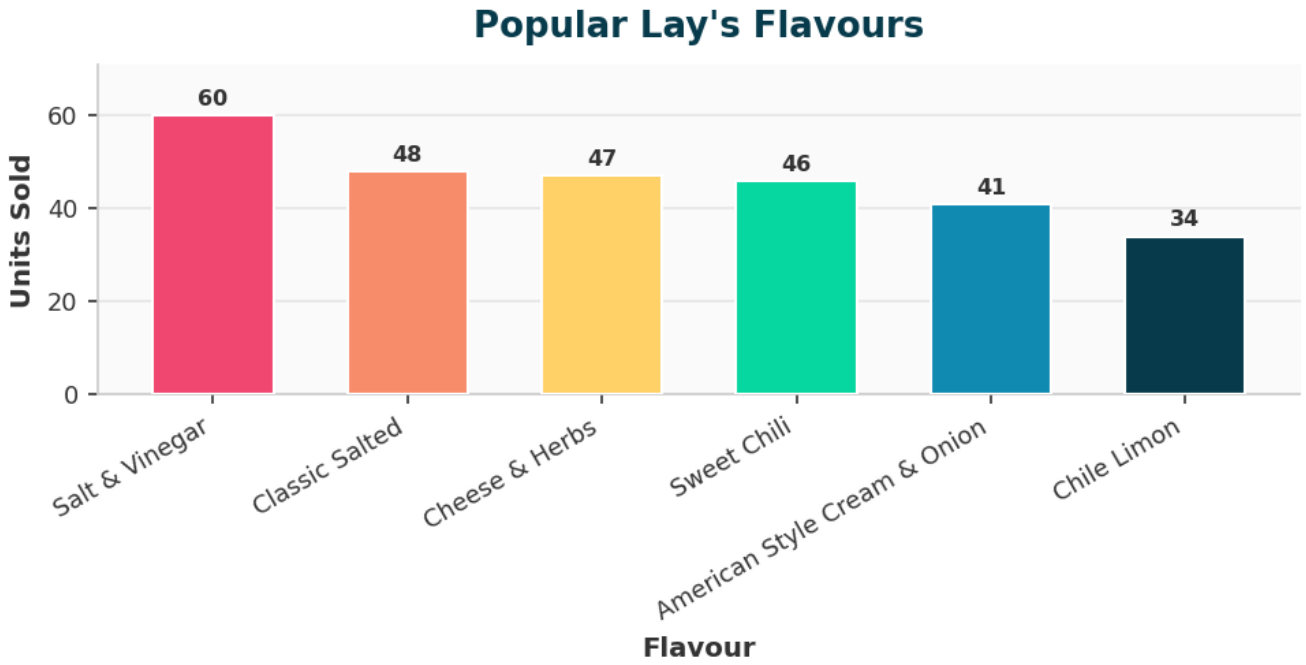
Additional Interpretation

Pack-size demand should be measured over longer real periods and adjusted for product availability before being used for procurement.

10. Flavour Preference

Figure Description

The figure compares simulated chips sales across the top individual flavours in the dataset. The X-axis represents flavour and the Y-axis represents Units_Sold. The chart provides a simple view of flavour preference within the academic dataset.



KPI

- Most popular flavour: Salt & Vinegar — 60 units
- Second-most popular flavour: Classic Salted — 48 units
- Flavours analysed: top 6

Observation

Salt & Vinegar is the most purchased simulated flavour, followed by Classic Salted. The difference in sales between flavours indicates that customers do not show equal demand for every flavour option.

Business Insight

Popular flavours can receive greater inventory allocation and online visibility. Lower-demand flavours can be stocked selectively or evaluated for regional or seasonal positioning. Flavour preferences can also be combined with pack size and category to identify high-demand product variants.

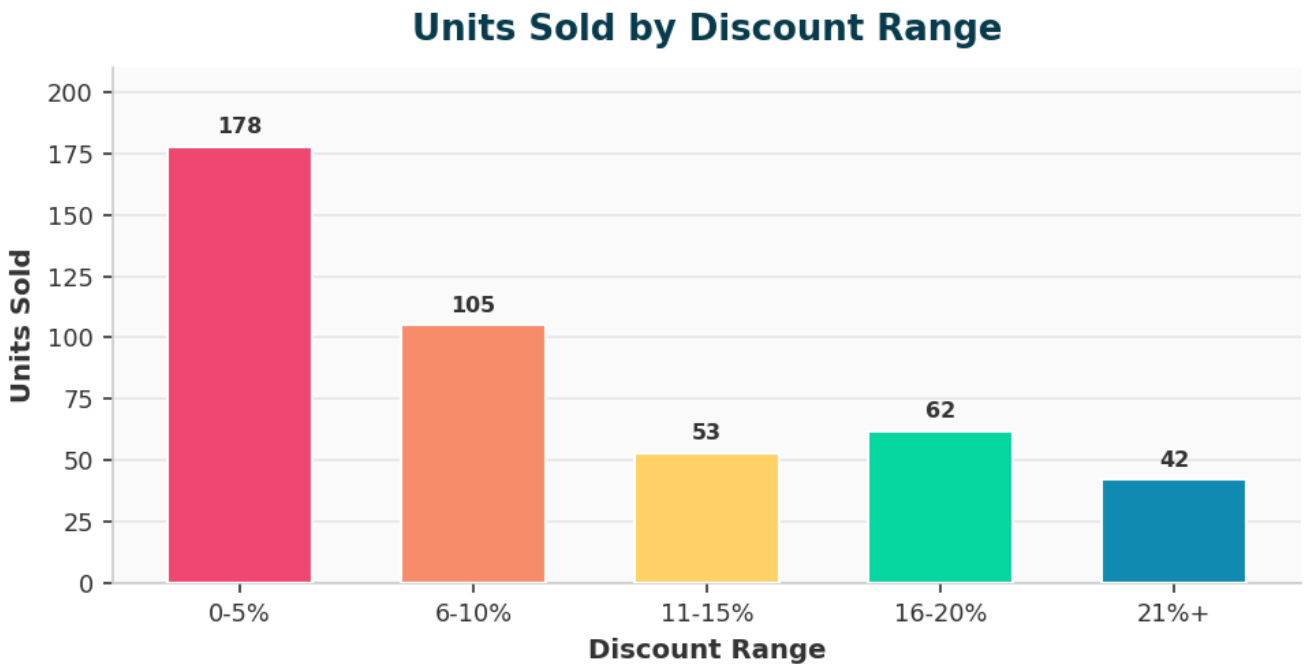
Additional Interpretation

Flavour popularity is synthetic and does not represent actual Lay's market preference.

11. Discount vs Sales

Figure Description

The figure compares observed sales across discount levels/ranges. The analysis is designed to determine whether higher-discount records are associated with different sales volumes. The chart should be interpreted carefully because many factors other than discount can influence sales, including product popularity, price, availability and customer demand.



KPI

- Average discount: 9.7%
- Highest discount observed: 25%
- Total simulated revenue: Rs. 24,729.5

Observation

Sales volume differs across discount ranges in the synthetic records. Some promotional levels show stronger observed unit sales, but this pattern alone cannot prove that discounting caused the additional sales. Product popularity and other factors may be influencing both variables.

Business Insight

Discount decisions should be evaluated using units sold, revenue and profit margin together. Businesses can use promotions strategically for selected flavours, clearance inventory or demand stimulation rather than applying high discounts universally. The objective should be profitable incremental demand, not simply higher unit volume.

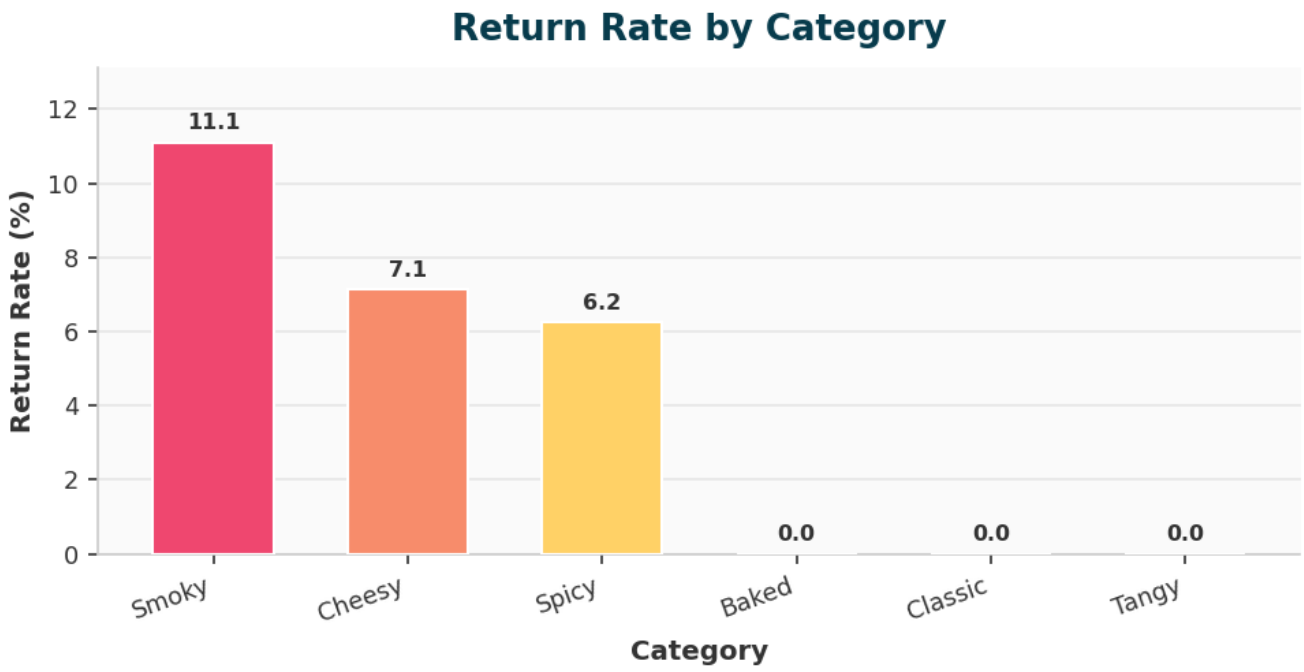
Additional Interpretation

The relationship shown is descriptive association, not a causal experiment.

12. Return Behaviour

Figure Description

The figure compares the percentage of simulated transactions marked as returned across flavour categories. The X-axis represents categories and the Y-axis represents the return rate. Return analysis is useful because high return levels may indicate the need to investigate packaging damage, freshness, quality or product-information issues.



KPI

- Overall simulated return rate: 4.0%
- Highest-return category: Smoky — 11.1%
- Lowest-return category: Tangy — 0.0%

Observation

The overall simulated return rate is 4.0%. Smoky has the highest category return rate, while Tangy has the lowest. The difference indicates that return behaviour varies by category within the synthetic records.

Business Insight

Categories with higher returns should be investigated using return reasons, packaging condition and customer feedback. Better handling during transit, clearer expiry information, quality checks and improved expectation-setting may reduce avoidable returns. Return analysis can therefore support both customer experience and cost control.

Additional Interpretation

A higher return rate does not automatically mean that a product is poor; the reason for the return must be examined.

Summary of Key Metrics

The following KPI table consolidates the main results of the synthetic dataset into a single management-style view. These metrics provide a quick summary while the preceding sections provide the detailed analytical explanation.

Objective	Key Metric	Result
Product performance	Best-selling product	Salt & Vinegar
Product performance	Best-selling product units	60
Daily sales	Average daily units	15
Daily sales	Highest-sales date	2026-07-05
Peak hours	Peak purchase hour	20:00
Category	Best-selling category	Spicy
Ratings	Highest-rated category	Cheesy (4.28/5)
Regional	Highest-sales city	Mumbai
Channel	Best channel by units	Offline/Store
Customer	Largest age group	45+
Pack size	Most popular pack size	30g
Flavour	Most popular flavour	Salt & Vinegar
Revenue	Total simulated revenue	Rs. 24,729.5
Sales volume	Total units sold	440
Customer experience	Overall average rating	4.09/5
Returns	Overall return rate	4.0%

KPI Interpretation

The KPI summary shows that product demand, customer satisfaction, geography, channel performance and return behaviour can all be measured from transaction-level data. In a real retail environment, these measures could be placed on a dashboard and monitored regularly to identify changes in demand and support faster decisions.

Important Academic Limitation

Every value in this KPI table is calculated from the synthetic dataset. It must not be presented as official Lay's / PepsiCo sales or customer statistics.

How This Would Work at Real Scale

At real scale, a snack-food analytics system could combine authorized sales data from physical stores and online channels with inventory, product, customer-service and return information. The descriptive framework used in this project could then be updated daily or in near real time.

Inventory Management

Monitor product, pack-size and flavour demand to reduce stock-outs and excess inventory. Store-level and warehouse-level availability can be compared with actual demand so that replenishment can be prioritized.

Product Planning

Identify categories and flavour variants with consistently strong demand. New flavour launches can be compared with previous products using units, revenue, ratings and return rates.

Regional Planning

Compare validated city and state demand to decide where inventory should be positioned. Local demand can be combined with flavour and pack-size preferences to improve assortment planning.

Marketing & Promotions

Evaluate campaigns by comparing units, revenue and customer response before, during and after promotional periods. Discount effectiveness should be measured with profitability rather than volume alone.

Customer Experience

Combine ratings, reviews and return reasons to identify possible issues with taste, freshness, packaging or product descriptions.

Online Sales

Use customer purchase and browsing behaviour to improve product recommendations, search ranking, product visibility and digital merchandising.

Management Dashboards

Create dashboards showing KPIs such as revenue, units, average rating, return rate, category contribution and channel performance so managers can monitor changes quickly.

Business Insight

A real-time analytics system would allow the business to move from simple historical reporting toward continuous decision support. High-demand flavours could be replenished faster, weak-performing variants could be reviewed earlier, and regional or channel differences could be addressed before they become larger operational problems.

Next Steps: Descriptive - Predictive Analytics

The descriptive analysis performed in this project provides a foundation for future predictive analytics. Descriptive analytics answers questions such as what sold the most, which category was popular and which channel generated more units. Predictive analytics can then use historical patterns to estimate what may happen next.

Possible Predictive Applications

- Forecast future chips sales by product, category and city using historical sales patterns.
- Predict demand for specific pack sizes and flavours to improve inventory planning.
- Identify products likely to become high-demand items before stock shortages occur.
- Predict customers who are likely to purchase again and support personalized offers.
- Estimate which products or orders have a higher probability of being returned.
- Optimize inventory allocation across stores, warehouses and online fulfilment locations.
- Recommend products based on previous purchases, category preferences, pack sizes and flavours.
- Estimate expected sales under different promotional or discount scenarios.

Example Business Scenario

If real data repeatedly showed strong demand for Salt & Vinegar in a particular city and pack size, a predictive model could estimate future demand for that product-pack-city combination. Inventory could then be positioned before demand occurs rather than only after stock begins to decline.

Business Insight

Moving from descriptive to predictive analytics can help a retailer become more proactive. Instead of only reporting past sales, the business could forecast demand, anticipate returns, personalize recommendations and optimize inventory. This could improve efficiency and reduce avoidable stock-outs and excess inventory.

Requirement for Real Implementation

Predictive models require larger, validated datasets, appropriate features, model evaluation and responsible handling of customer information. The current synthetic dataset is suitable for demonstrating descriptive analysis but is not sufficient to train a reliable production forecasting model.

References

1. LAY'S / PepsiCo Official Website. https://www.lays.com/
2. LAY'S Official product and brand information.
3. Python Documentation. https://docs.python.org/
4. Pandas Documentation. https://pandas.pydata.org/docs/
5. Matplotlib Documentation. https://matplotlib.org/stable/
6. Kaggle — dataset and analytics resources. https://www.kaggle.com/
7. Microsoft Excel documentation and spreadsheet analysis resources.

Dataset Citation

Synthetic LAY'S Chips Sales Dataset (150 transaction records), generated for academic descriptive analytics purposes using simulated snack-food sales, customer preferences, product attributes, Indian market locations and realistic retail-style variables.

Source-use Note

Official brand sources are used only for general brand/product context. All transaction-level sales, customer, revenue, rating, return and KPI results in this report are generated from the synthetic academic dataset.

Appendix

Appendix A — Sample of Synthetic Dataset

```
Product_ID | Product_Name | Category | Pack_Size | Price | Discount | Units_Sold | Revenue | City | Channel
LY-P02 | Magic Masala | Spicy | 90g | 50 | 20 | 4 | 160.0 | Chandigarh | Offline/Store
LY-P02 | Magic Masala | Spicy | 52g | 30 | 25 | 4 | 90.0 | Hyderabad | Online
LY-P04 | American Style Cream & Onion | Tangy | Family Pack (200g) | 110 | 5 | 4 | 418.0 | Delhi | Online
LY-P01 | Classic Salted | Classic | 150g | 80 | 15 | 2 | 136.0 | Mumbai | Online
LY-P04 | American Style Cream & Onion | Tangy | 150g | 80 | 5 | 2 | 152.0 | Lucknow | Offline/Store
LY-P05 | Spanish Tomato Tango | Spicy | 150g | 80 | 0 | 5 | 400.0 | Chandigarh | Offline/Store
LY-P07 | Sweet Chili | Spicy | 30g | 20 | 0 | 3 | 60.0 | Chandigarh | Offline/Store
LY-P07 | Sweet Chili | Spicy | 52g | 30 | 0 | 2 | 60.0 | Delhi | Online
LY-P12 | Salt & Vinegar | Classic | 150g | 80 | 10 | 2 | 144.0 | Lucknow | Online
LY-P01 | Classic Salted | Classic | 150g | 80 | 10 | 4 | 288.0 | Bengaluru | Online
```

Appendix B — Python / Pandas Data Preparation

```
import pandas as pd
df = pd.read_csv('lays_synthetic_150.csv')
df.drop_duplicates(inplace=True)
df['Purchase_Date'] = pd.to_datetime(df['Purchase_Date'])
df['Final_Price'] = df['Price'] * (1 - df['Discount']/100)
df['Revenue'] = df['Final_Price'] * df['Units_Sold']
df['Age_Group'] = pd.cut(df['Customer_Age'], [0,17,24,34,44,100], labels=['Under 18','18-24','25-34','35-44','45+'])
```

Appendix C — KPI Calculation Examples

```
Total Units = df['Units_Sold'].sum()
Total Revenue = df['Revenue'].sum()
Average Rating = df['Customer_Rating'].mean()
Best Product = df.groupby('Product_Name')['Units_Sold'].sum().idxmax()
Best Category = df.groupby('Category')['Units_Sold'].sum().idxmax()
Best City = df.groupby('City')['Units_Sold'].sum().idxmax()
Return Rate = (df['Return_Status'].eq('Yes').mean()) * 100
```

Appendix D — Visualization Examples

```
import matplotlib.pyplot as plt
product_sales = df.groupby('Product_Name')['Units_Sold'].sum().sort_values(ascending=False)
product_sales.head(8).plot(kind='bar')
plt.title("Top LAY'S Products by Units Sold")
plt.xlabel('Product'); plt.ylabel('Units Sold'); plt.show()
```

Appendix E — Supporting Analysis Tables

Supporting tables can be generated from the same dataset using Pandas groupby operations for product, category, city, channel, age group, pack size, flavour, discount and return status. These tables can be exported to Excel for further academic verification.

Appendix F — Academic Limitation

The dataset is synthetic and therefore cannot represent actual Lay's / PepsiCo sales, customers, revenue, market share or official product performance. The purpose of the appendix is to make the analysis reproducible and demonstrate the use of Python, Pandas and Matplotlib in a descriptive analytics workflow.